SCADA-Based Solar Tracker System

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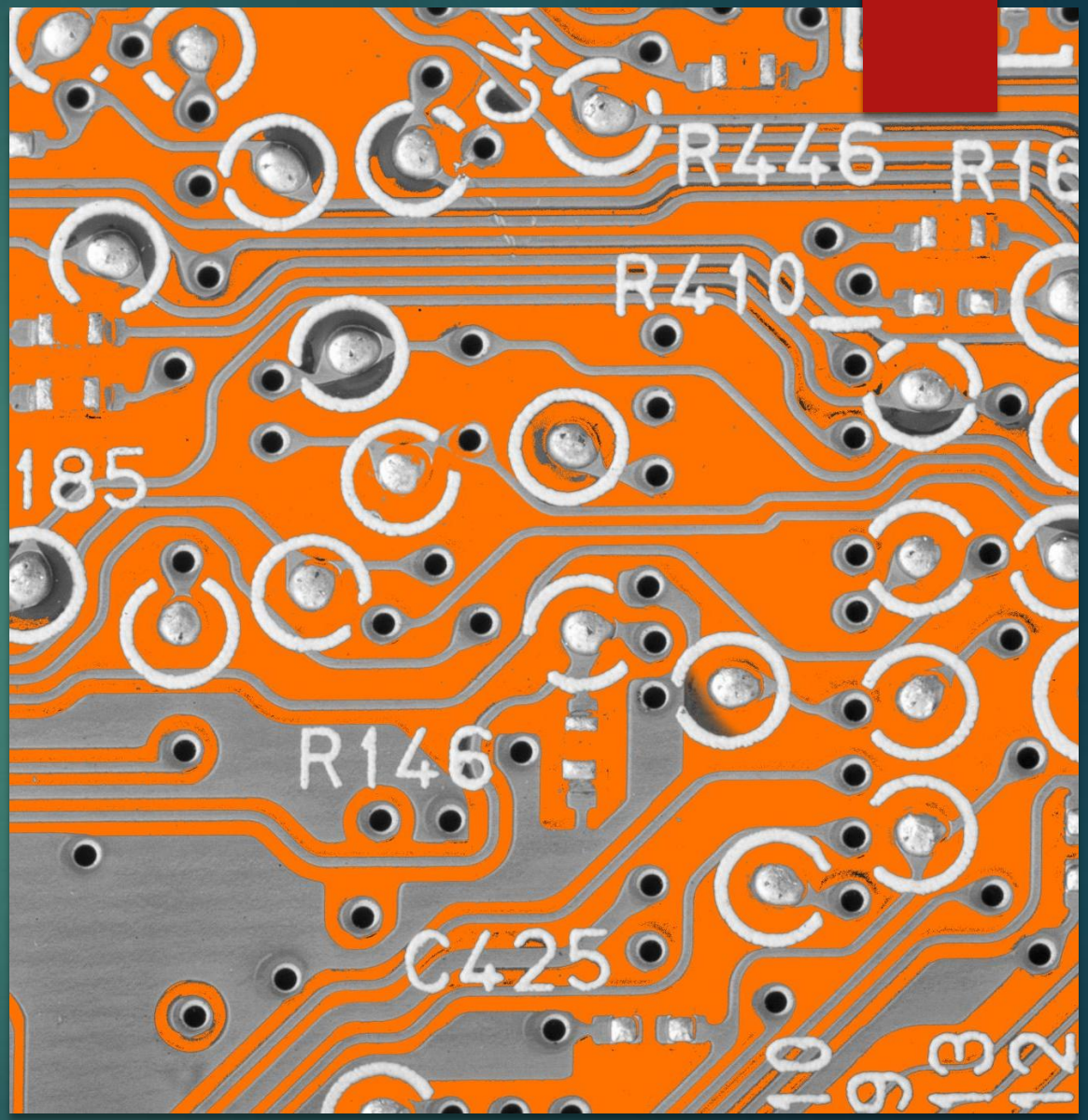
Objectives

Design	• Design a solar tracker system
Track	• Track sun position automatically
Increase	• Increase solar panel efficiency
Monitor and control	• Monitor and control system using SCADA
Combine	• Combine mechanical, electrical, and programming skills

System Overview

- **System has 4 main parts:**
- Sensors (LDR)
- Controller (Arduino)
- Motor (Servo)
- SCADA Interface (Computer)



Main Components

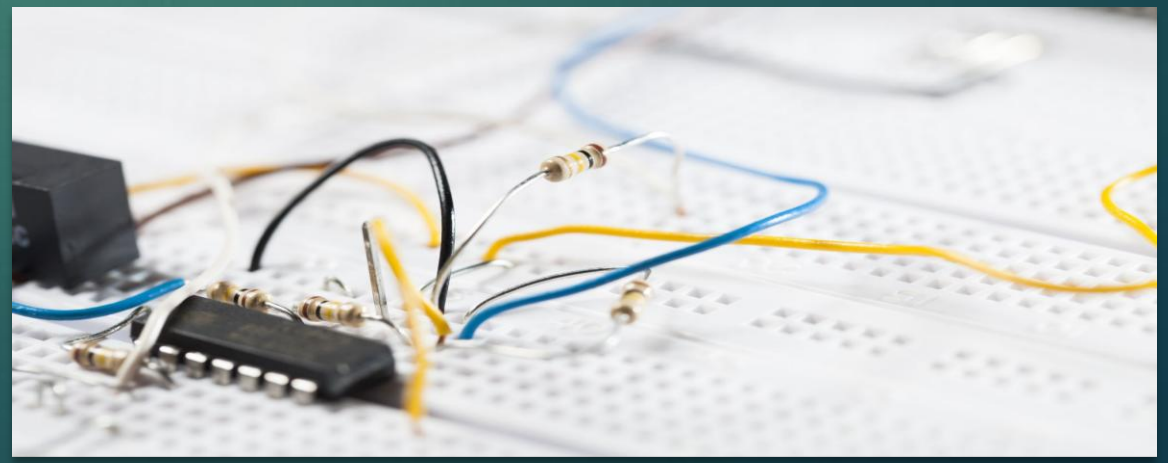
Component	Function
Arduino	Control system
LDR Sensors	Detect sunlight
Servo Motor	Moves panel
L293D	Motor driver
Solar Panel	Generates power
Bracket	Holds panel
SCADA	Monitoring

Electrical System

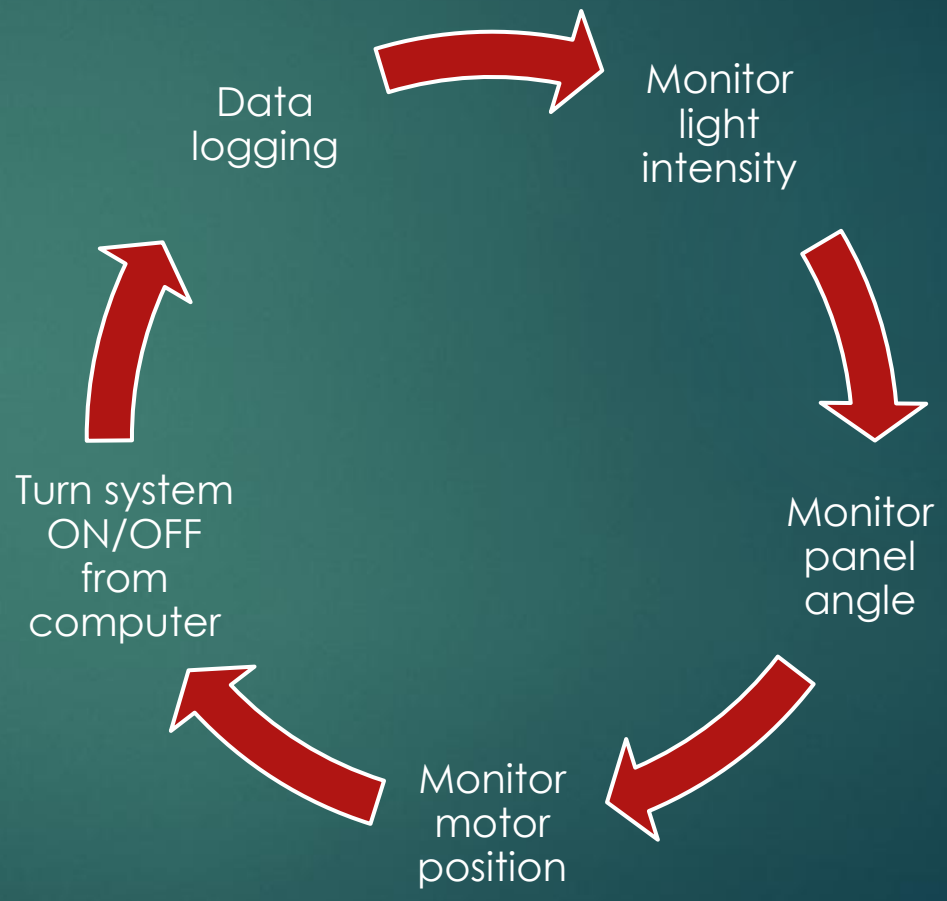
- LDR voltage divider
- Arduino connections
- Motor driver
- Power supply

Control Logic (Arduino Code)

- If left sensor > right sensor → move left
  If right sensor > left sensor → move right
  If equal → stop



SCADA System



Software Used

Software	Purpose
SolidWorks	Mechanical design
Arduino IDE	Programming
Tinkercad	Circuit simulation
MATLAB	Data analysis
SCADA	Monitoring
Excel	Data recording

Conclusion

- Built a working solar tracking system
- Integrated mechanical + electrical + software
- Learned SCADA and automation
- Real-world engineering project